

# Backend Development Summary

Kemgang Prince (Leprince) • Backend Developer

01

## 01 PROJECT AT A GLANCE

A practical digital response to forged titles, overlapping claims, and difficult land-record checks.

LandVerifyCM is a secure web-and-mobile land verification platform for Cameroon. Citizens can search active parcels by title number, browse by quarter, and inspect ownership, GPS, status, and documents. Administrators maintain the registry through controlled workflows. The backend bridges the React/Expo interfaces and PostgreSQL: it applies rules, protects operations, and returns consistent JSON.

## 02 MY CONCRETE RESPONSIBILITIES

- **REST API** — organized authentication, public land, and administrator routes.
- **Business logic** — implemented controllers for accounts, parcels, ownership transfers, users, and audit history.
- **Verification** — connected title/GPS checks to record creation and updates.
- **Integration** — served stable responses consumed by the browser and mobile application.

## 03 MAIN API SURFACES

- **Public** — register, login, password recovery, title search, quarter browsing, parcel details, and quarter lists.
- **Admin** — list, upload, edit, deactivate, attach documents, manage ownership records, inspect users, and read audit logs.
- **Operations** — startup checks, protected uploads, health monitoring, and error handling.

## 04 FROM REQUEST TO VERIFIED RECORD

Each layer has one clear responsibility, which keeps the system easier to test, secure, and extend.

| 1                   | 2                    | 3                       | 4                      | 5                    |
|---------------------|----------------------|-------------------------|------------------------|----------------------|
| <b>WEB / MOBILE</b> | <b>EXPRESS ROUTE</b> | <b>MIDDLEWARE</b>       | <b>CONTROLLER</b>      | <b>PRISMA + SQL</b>  |
| sends HTTPS request | selects endpoint     | checks input and access | applies business rules | reads or writes data |

Routes separate `/api/auth`, `/api/lands`, and `/api/admin`. Controllers handle requests, then Prisma sends typed queries to PostgreSQL. Pagination controls lists; selected fields protect passwords. Parcel details join documents, uploader information, and ordered ownership history.

## VERIFICATION DECISION RULES

- 1 Matching title number → **DUPLICATE**
- 2 GPS distance under 10 m → **DUPLICATE**
- 3 GPS distance under 50 m → **SUSPICIOUS**
- 4 No conflict → **VALID**

## 05 WHY THE ALGORITHM MATTERS

The Haversine formula converts GPS pairs into distance using Earth's radius (6,371 km). Checks compare active parcels and exclude the current parcel during updates. A duplicate upload returns HTTP 409; a nearby record is stored as Suspicious for manual review.

**DESIGN VALUE** One verification service means the browser and phone receive the same evidence-based decision.

# Built for Safe Public Use

02

Security controls protect the registry while the technology keeps development practical.

## 06 SECURITY AND ACCOUNTABILITY

- **Identity** — bcrypt uses 12 rounds; JWT carries user ID and role.
- **Access** — Bearer authentication and ADMIN authorization protect management and files.
- **Abuse controls** — Helmet, CORS, body limits, and rate limits reduce attacks.
- **Evidence** — important changes receive timestamped audit records.
- **Uploads** — Multer permits five PDF/JPG/PNG files, maximum 10 MB each.

## 07 TECHNOLOGIES AND WHY

- **TypeScript** — catches data-shape mistakes before deployment.
- **Node.js + Express** — delivers a lightweight REST service.
- **PostgreSQL + Prisma** — provides relational storage, constraints, migrations, and typed queries.
- **Zod** — validates coordinates, years, areas, and ownership data.
- **JWT, bcrypt, Multer** — support sessions, safe passwords, and controlled uploads.

## 08 HOW THIS HELPS PEOPLE IN CAMEROON

The backend turns scattered land information into checks citizens and administrators can act on.

### BUYERS

Check a title and location before paying, reducing exposure to forged or overlapping claims.

### FAMILIES & PROFESSIONALS

Follow ownership history and supporting evidence when investigating a parcel.

### ADMINISTRATORS

Maintain records with controlled changes, traceable actions, and clear manual-review flags.

**PRINCE'S CONTRIBUTION** A consistent and secure backend that converts land records into useful verification results for both web and mobile users.